

Design and Analysis Process for AIDA mission

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PART1

Introduction:

mission statement

The Space Mission Analysis and Design:

Define Objectives & Characterize the Mission

발표자: 강석주

1. Mission Statement

1. Mission Statement

- NEOs (Near Earth Objects)



2.1 Define broad objectives & Constraints

Mission Objectives	
Primary Objectives	소행성 충돌 실험으로 소행성의 공전 궤도의 변화를 detect한다.
Secondary Objectives	소행성을 가까이에서 관찰하며 충돌 시 섬광을 분광 관측하여 chemical composition을 분석한다.

2.2 Estimate quantitative mission needs and requirements

- 소행성 65803 Didymos secondary



2.2 Estimate quantitative mission needs and requirements

Requirements	
Orbit accuracy	소행성의 지름이 약 150m이므로 center of mass에서 25m의 정확도를 만족해야 한다.
S/C 개수	1개 또는 2개
궤도 분석 오차	주기 변화율 10% 이내
Mission life time	3 years

2.3 Define alternative mission concepts

- 궤도: Swingby 이용하여 rendezvous

Concepts		
충돌 방법	Impactor	Impact probe
S/C 개수	1개	2개
Mission controlling	Autonomous + ground command	Ground command

2.3 Define alternative mission concepts

- Data concepts

Data concept		
Data delivering	Direct downlink	Store & downlink
궤도 분석	Radar	Laser
정보 분석	적외선	가시광선

2.4 Define alternative mission architectures

- 발사장소

the Centre Spatial Guyanais in French Guyana

- 발사체

VEGA



2.4 Define alternative mission architectures

1st architecture

Concepts		
충돌 방법	Impactor	<u>Impact probe</u>
S/C 개수	1개	<u>2개</u>
관찰 위성이 소행성에 대한 data를 얻은 후에 충돌 위성이 소행성에 충돌하고 그 후의 궤도 변화를 계산. 그 후 관측위성은 지구에 돌아온다.		

2.4 Define alternative mission architectures

2nd architecture

Concepts		
충돌 방법	<u>Impactor</u>	Impact probe
S/C 개수	<u>1개</u>	2개
모위성이 impactor를 가지고 소행성과 랑데뷰하고 소행성에 impactor를 발사한다. 충격을 준 후에 궤도의 변화를 관찰한다.		

2.5 Identify system drivers for each

- S/C 개수
(2대 이용하면 상대적으로 비용 감소)
- 충돌 방법
(위성 개수를 정하는 drivers)

2.6 Characterize mission concepts and architectures

- **Data delivering**
:Store & downlink
- **Payload**
:radar, image camera, 분광관측기, 열적외선 camera
- **명령**
:Autonomous + ground command
- **정보분석**
:적외선 카메라와 가시광선 카메라
- **추력기**
:chemical propulsion, Solar Electric Propulsion(SEP)

PART2

The Space Mission Analysis and Design: Evaluate the Mission & Define Requirements

발표자: 이영로

2.7 Identify critical requirements

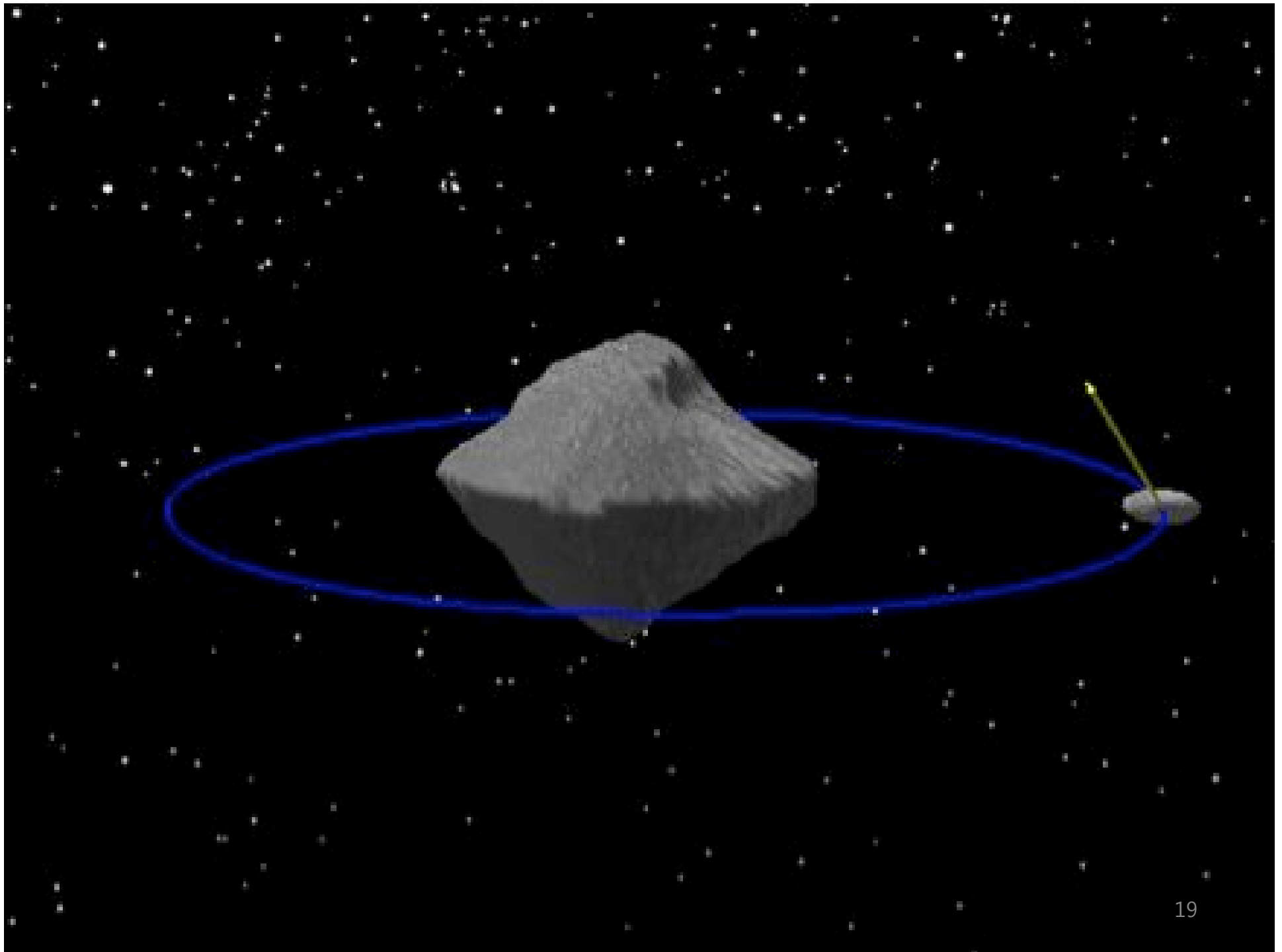
mission 의 성공유무를 결정하는 중요한 요인,
cost결정요인

- 소행성 결정
- 정확한 maneuver, 정확한 충격
(자세제어)

2.8 Evaluate mission utility

design, cost 측면에서
requirements, constraints를 얼마나 잘 충족시
키는가

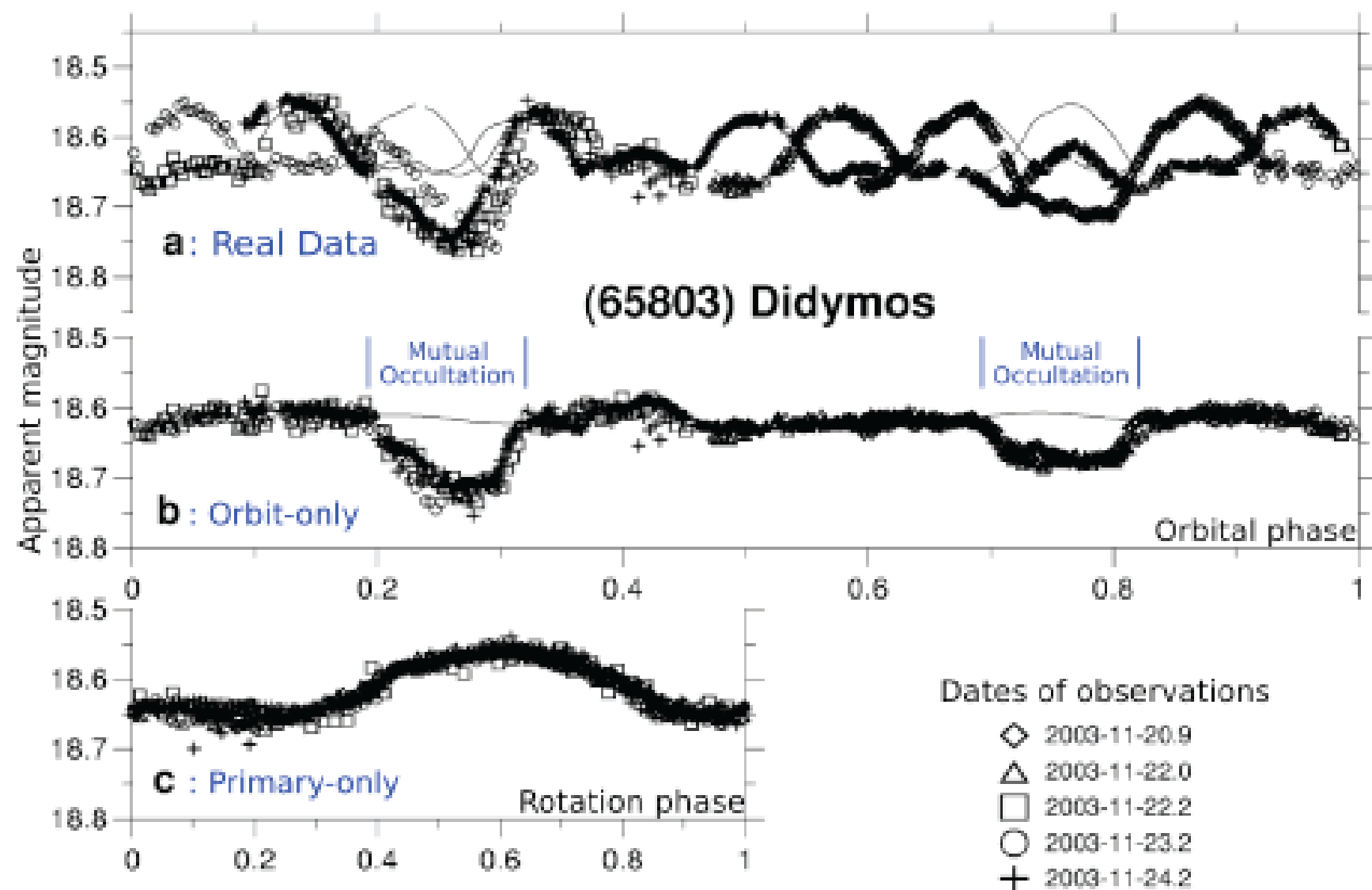
- independent한 mission : DART 와 AIM
- Low cost



Binary near-Earth asteroid (65803) Didymos (1996 GT)

Semi-major axis	1.644 AU
Eccentricity	0.384
Inclination	3.4°
Perihelion	1.01 AU
Aphelion	2.28 AU
Heliocentric period	770.14 days
Primary rotation period	2.26 hr
Binary orbit period	11.91 hr
Binary orbit semi-major axis	1.05 km
Primary diameter	800 m
Secondary diameter	150 m
Separation	1100 m
Pole Solutions	$(\lambda, \beta) = (157^\circ, 19^\circ); (329^\circ, -70^\circ)$
Spectral Type	Xk
H Magnitude	18
Geometric albedo	0.147

Didymos 의 Lightcurves



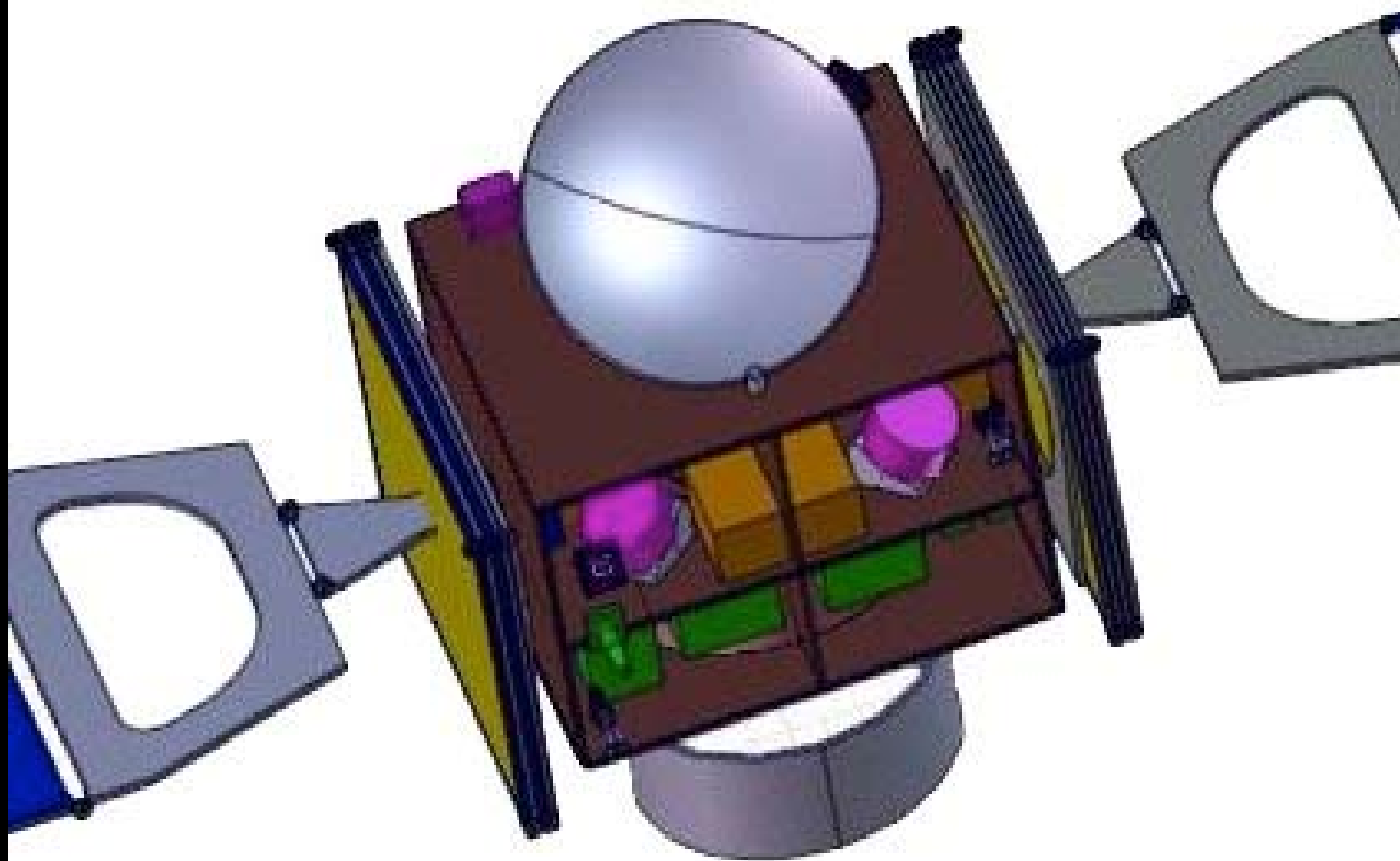
Didymos를 관측 할 수 있는 기회들

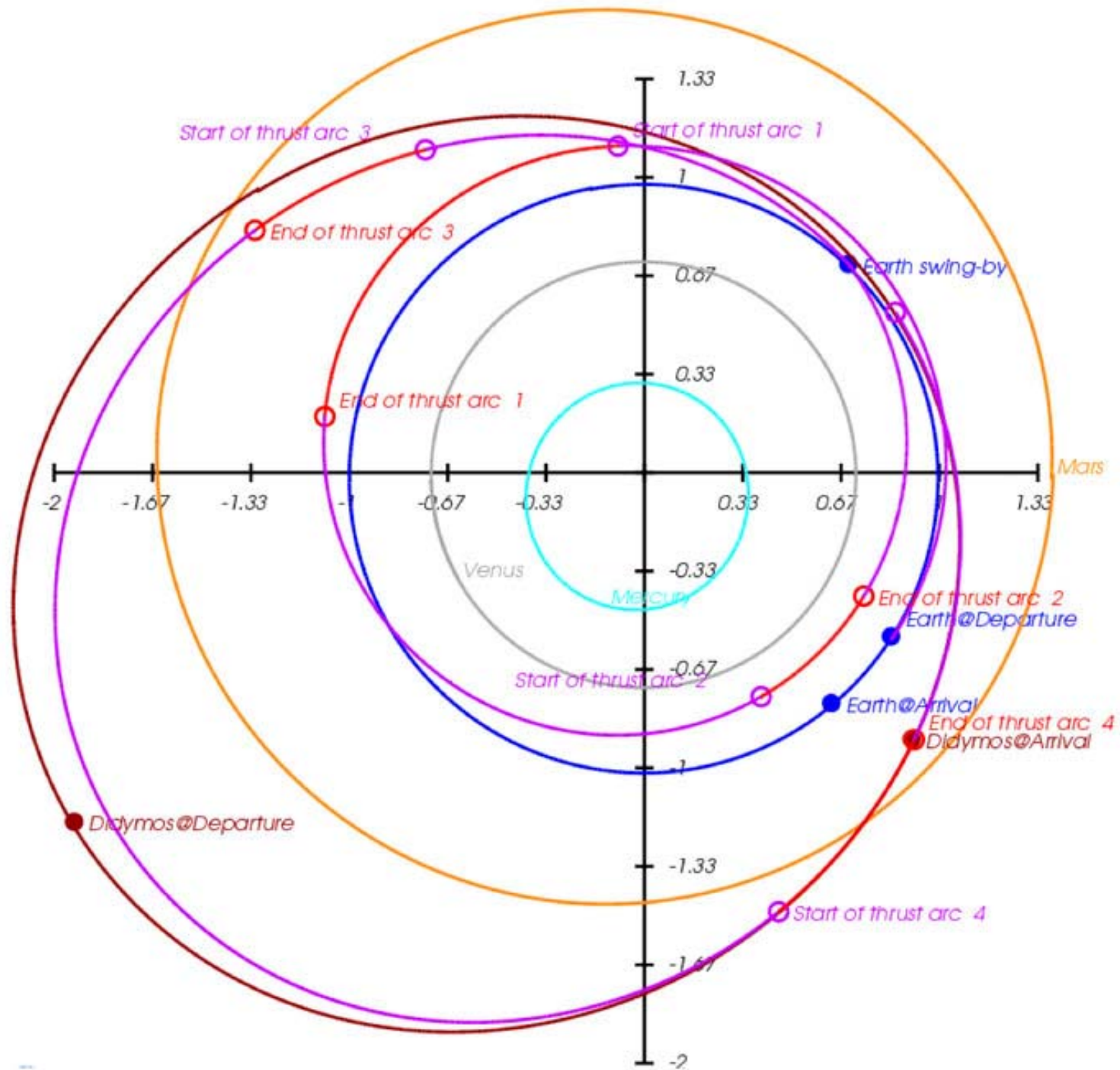
Observing Opportunity	Mag Range	Pole 1 Earth	Pole 1 Sun	Pole 2 Earth	Pole 2 Sun
2013 Feb - 2013 Jul	21-22	2013 Feb - 2013 Mar	2013 Jun - 2013 Aug	All	All
2015 Jan - 2015 Jun	21-22	none	none	All	All
2017 Jan - 2017 Jun	21-22	none	none	All	All
2019 Jan - 2019 Jun	20-22	none	none	All	All
2020 Dec - 2021 May	19-22	none	none	2020 Dec - 2021 Jan; 2021 Apr - 2021 May	All
2022 Apr - 2023 Apr	14-22	2022 Oct	2022 Apr - 2022 May; 2022 Nov - 2022 Dec	2022 Apr - 2022 Jun; 2022 Oct - 2023 Apr	All
2024 Apr - 2024 Sep	19-22	2024 Jul - 2024 Sep	2024 Apr - 2024 Jun	All	All
2026 Apr - 2026 Aug	20-22	All	2026 Apr - 2026 Jul	All	All

Facility	Use	Notes
Small (1-2 m) telescopes	Light-curves at some times, imaging	National facilities/University facilities
NASA IRTF (3 m)	Light-curves, spectroscopy, imaging	Goal to support NASA missions
Apache Point Observatory	Light-curves, imaging	JHU/APL has (small) share
Keck Observatory (10 m)	Imaging	NASA has small share
Hubble Space Telescope	Light-curve? Imaging	Probably defunct by impact
ALMA	Sub-mm imaging	Might separate components
Spitzer	Imaging, physical properties	
JWST		Cannot track NEOs
WISE	Physical properties	"Might get in 2015". Defunct by 2018. Not pointable.
Gaia	Astrometry	ESA mission
Pan-STARRS/LSST	Imaging?	Unclear if pointable. Not obviously useful.
Radar	Imaging, detection	S/N>30 only Aug-Dec 2022
Gemini AO	Imaging	Maybe separate components?
Magdalena Ridge Observatory	Light-curves, imaging	Specialized in NEO light-curves.

AIM (Asteroid Impact Monitoring)







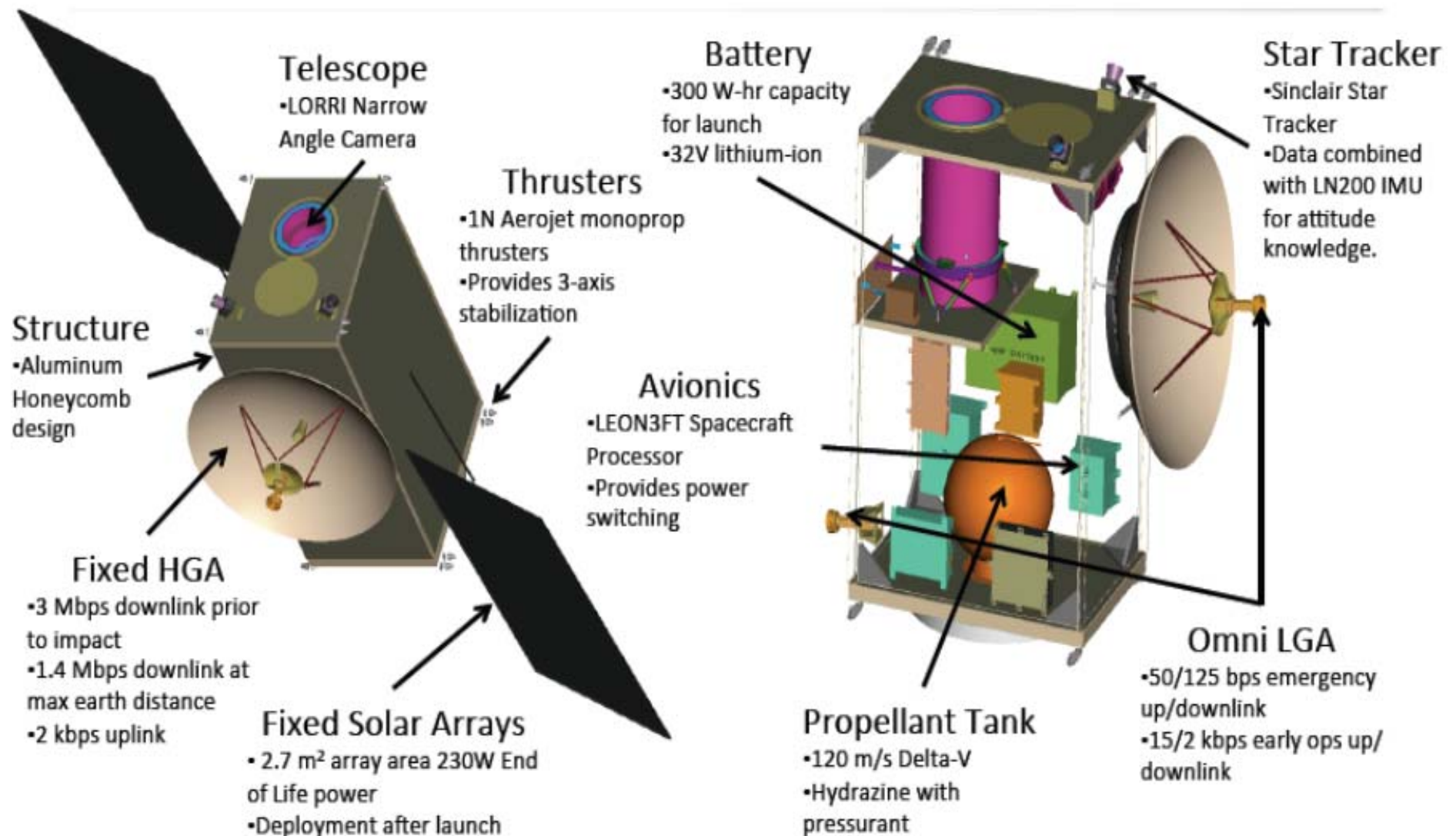


AIDA

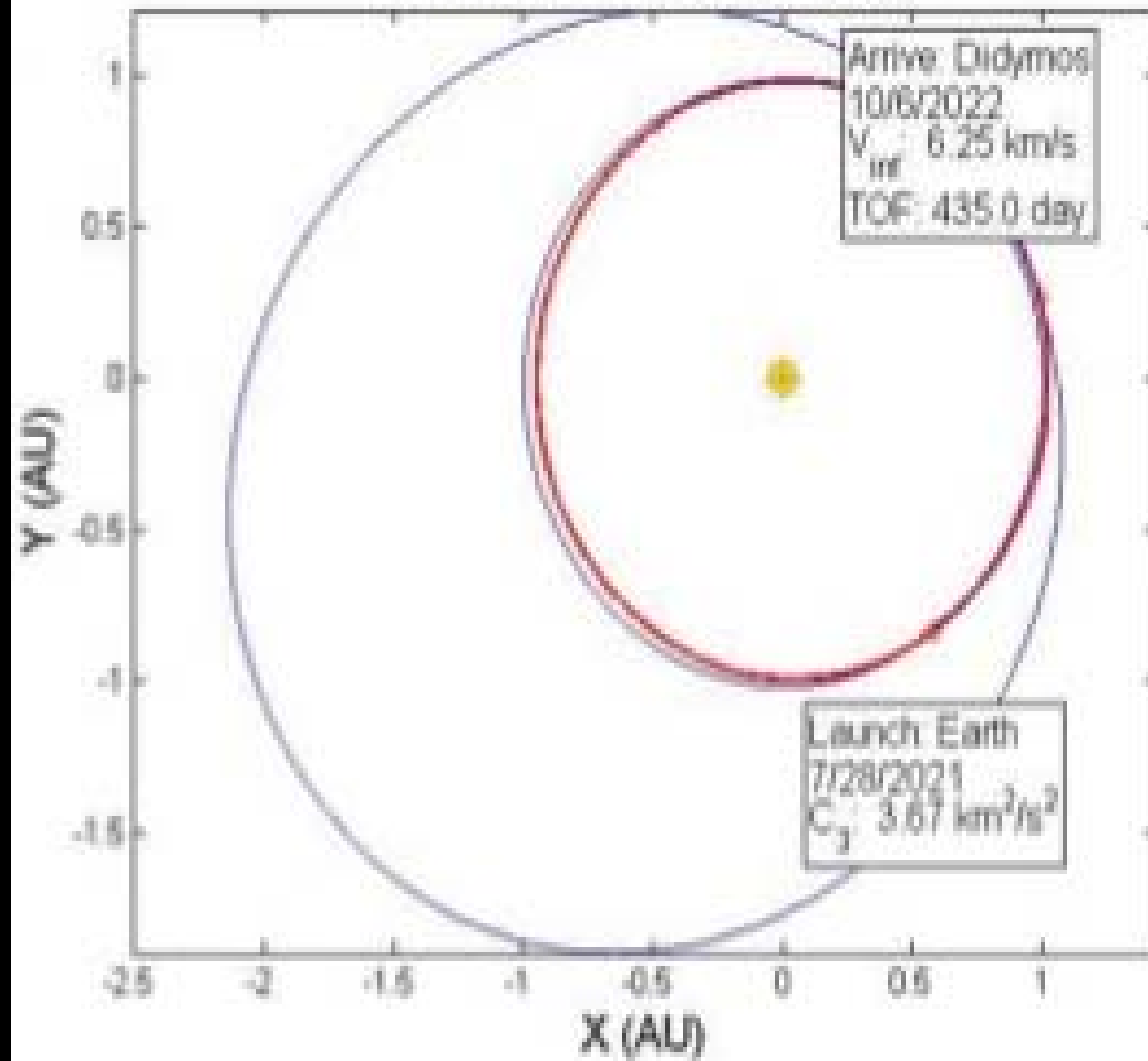
DART

(Double Asteroid Redirection Test)





Interceptor Primary Trajectory





2.10 Define system requirements



2.11 Allocate requirements to system elements

	AIM	DART
payload	소행성의 궤도 및 지질학적 특성을 조사하기 위한 카메라를 장착해야 한다. 충돌영상을 분석할 수 있는 장비를 실어야 한다.	자동항법장치를 위한 카메라를 실어야 한다
subsystem		
ADCS (자세결정및제어계)	정밀한 자세제어가 필요하다.	정밀한 자세제어가 필요하다.
TTC (통신계)	소행성 관측 데이터를 지상국으로 송신해줘야 한다.	자신의 궤도를 지상국으로 송신 해줘야 한다.
CDH (명령계)	적절한 순간에 적절한 mode 를 사용해야 한다.	적절한 순간에 적절한 mode 를 사용해야 한다.
EPS (전력계)	많은 종류의 payload 를 사용하고, SEP 를 사용하므로 효율성이 좋은 배터리를 사용하여야 한다.	리튬이온 전지를 이용한다.
TCS (열제어계)	소행성궤도에서 온도는 150~400K까지 온도차가 크다	소행성궤도에서 온도는 150~400K까지 온도차가 크다
SMS (구조계)	launcher 에 들어가는 box 형	소행성과 직접 부딪히므로 튼튼해야 한다.
GNC (항법유도제어계)	자신의 위치를 지상국에게 알린다.	자동항법장치로 스스로 소행성과 부딪혀야 한다.
PS (추진계)	소행성 궤도까지 먼 거리를 가므로 효율이 좋은 엔진을 사용하여야 한다.	chemical thrust 를 사용한다

AIM의 payload

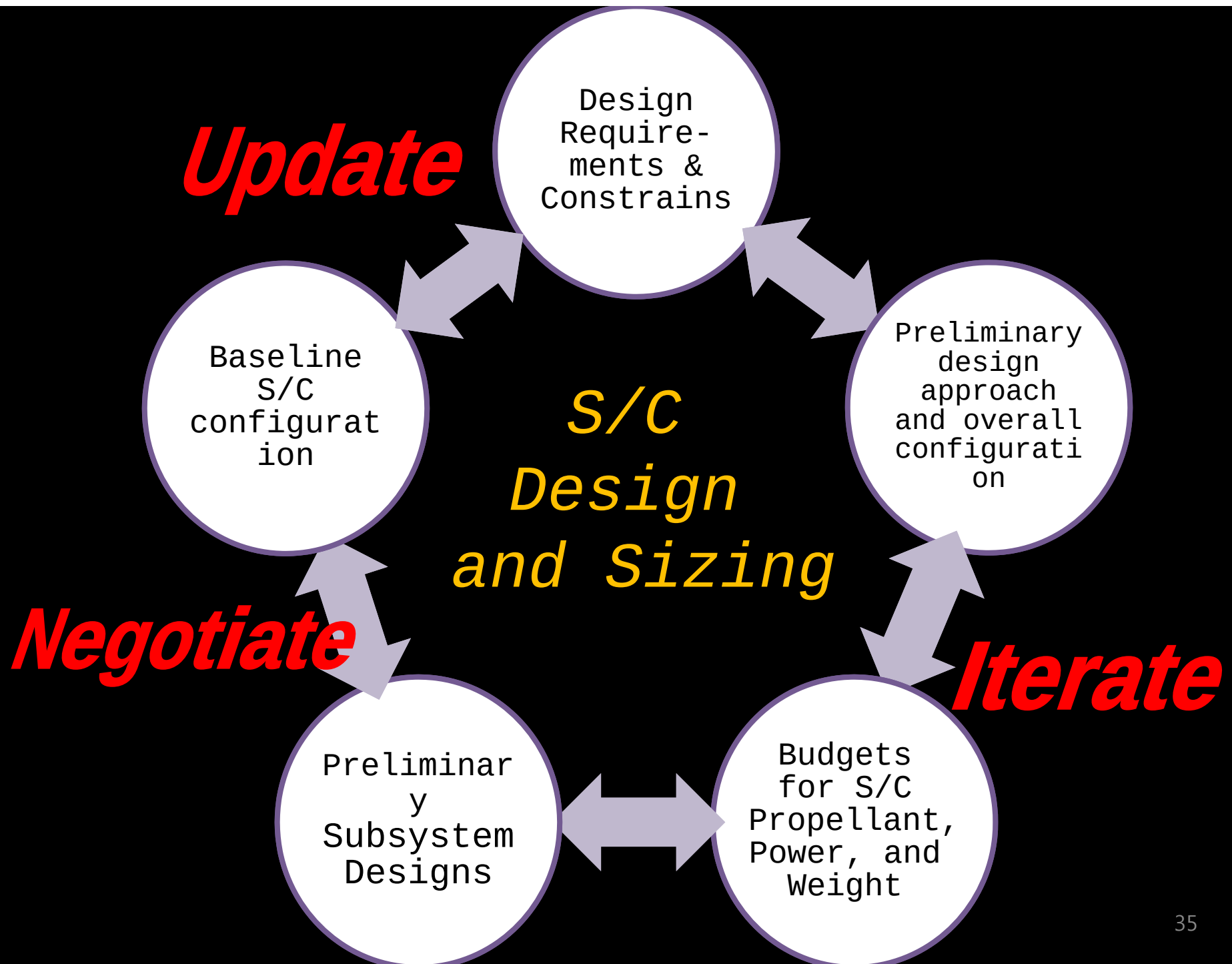
Instrument	Mass (kg)	Power (W)	Data Rate (kbps)	FOV (°)	Aperture (mm)	Length (mm)	Width (mm)	Height (mm)
<i>Narrow Angle Camera</i>	1.5	0.75	1	4	20	200	150	50
<i>Micro Laser Altimeter</i>	2.5	4.0	0.5	0.003	30	200	150	50
<i>Thermal IR Imager</i>	1.5	1.0	0.1	4	40	60	40	40
<i>NIR spectrometer</i>	1.5	7.0	10	4	38	100	50	50

PART3

The Spacecraft Design and Sizing:

- ❖ Principal requirements and constraints
- ❖ Preliminary spacecraft design approach and overall configuration
- ❖ Budgets for spacecraft propellant, power, and weight
- ❖ Preliminary subsystem designs

발표자: 이정아



3.1 Design Requirements and Constrains

AIM (rendezvous probe)

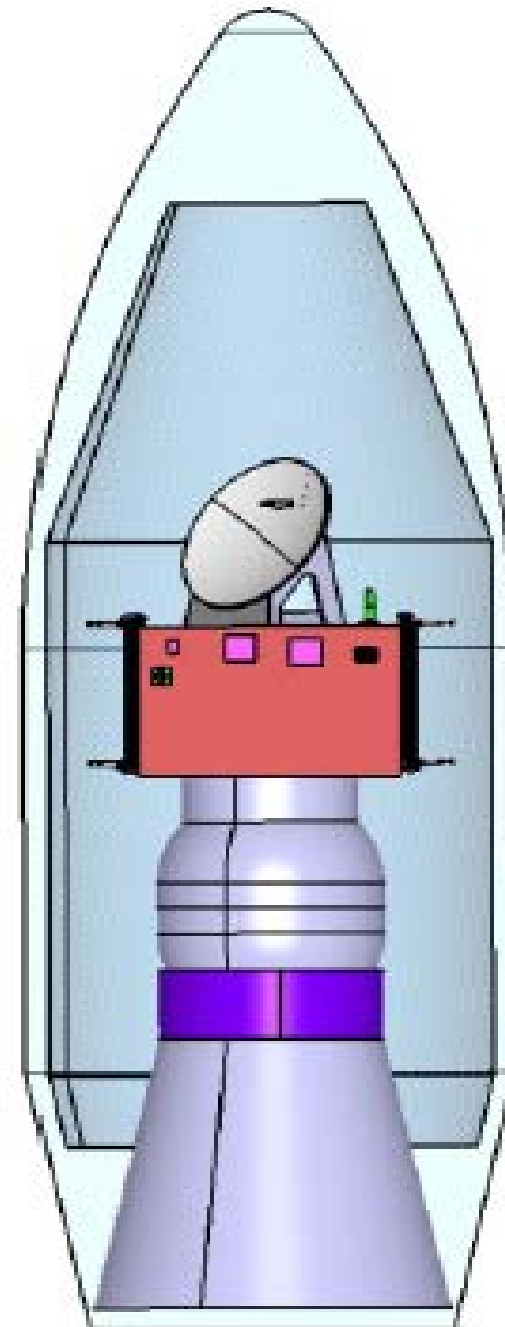
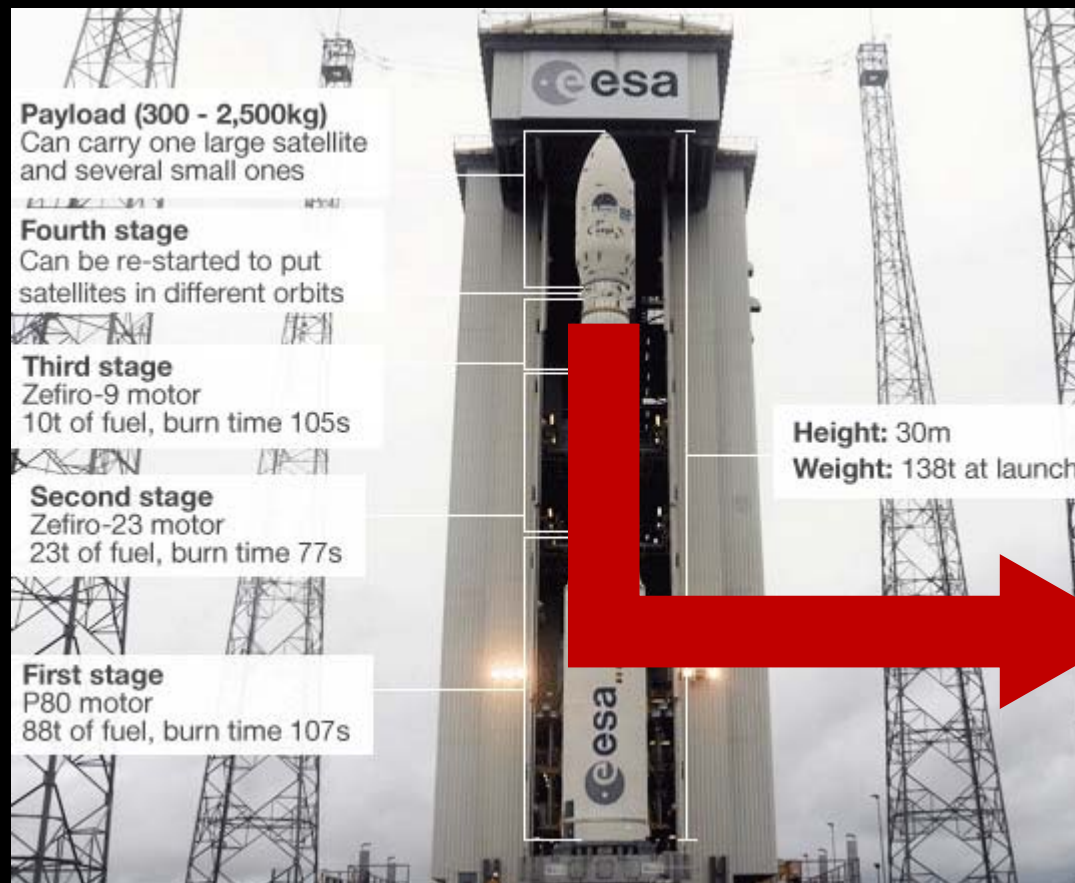
Mission	Payload	Orbit	Environ-ment	Launch System	Ground System
Asteroid Impact Monitoring	Imagers, Radar, Spectrometer, & Antenna	13.5-17km/100km from the asteroid	VEGA launch; 2/3 months before and after impact	VEGA-LV with kick motor; Guiana Space Centre	At least 2 stations in different hemispheres

DART (impact probe)

Mission	Payload	Orbit	Environ-Ment	Launch System	Ground System
Asteroid system Redirect ion	Optical imager, Radar & Antenna	1AU from Sun, <0.11AU with earth	Minotaur V launch; 435days and then impact	Minotaur V; Vandenberg Air Force Base	At least 2 stations in different hemispheres

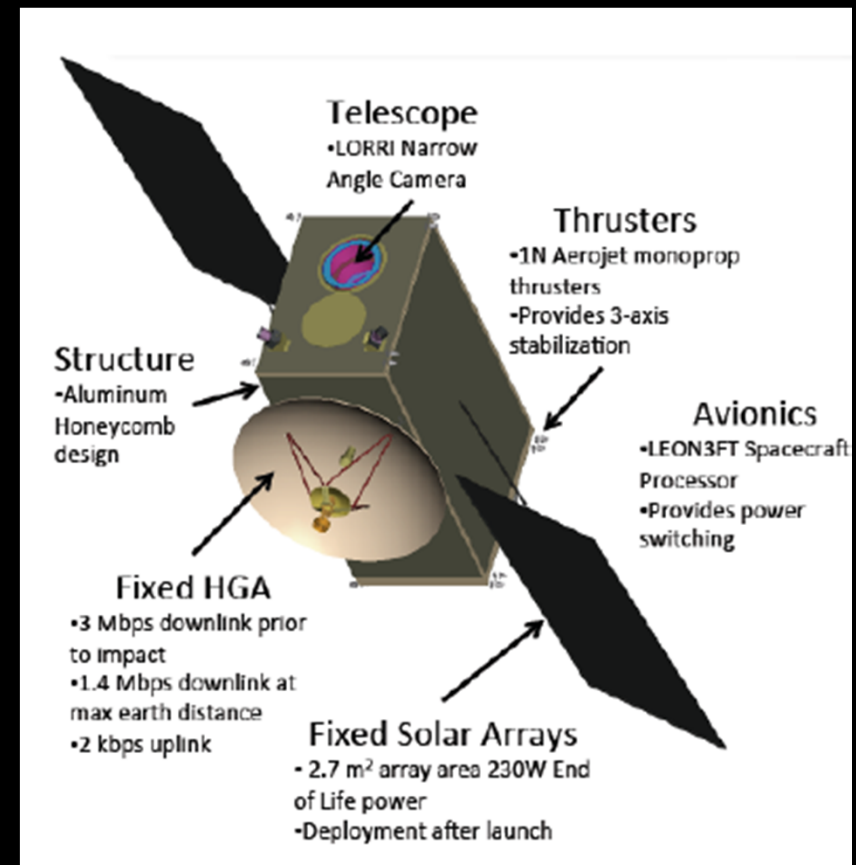
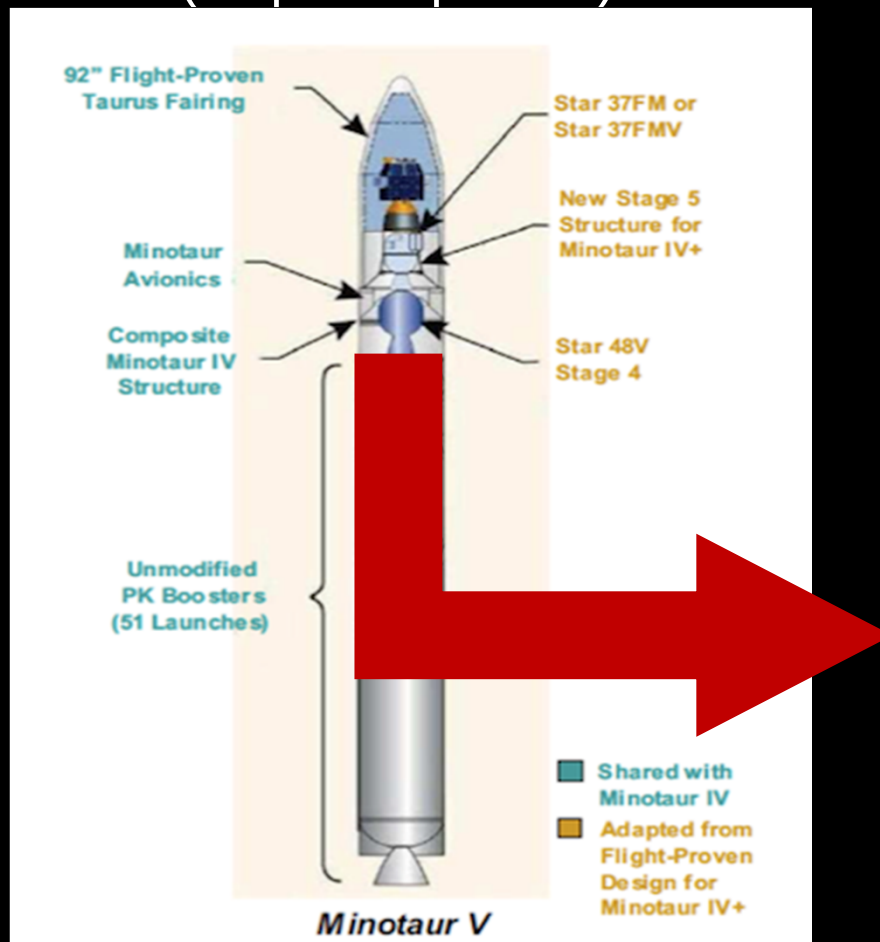
3.2 S/C Design And Overall Configuration

AIM (rendezvous probe)



3.2 S/C Design Approach & Overall Configuration

DART (impact probe)



3.3 Budgets for S/C Propellant

AIM (rendezvous probe)

Element	Mass(kg)
Velocity Control	
- Orbit insertion	40
- Error correction	2
- Stationkeeping	7
- Station Change	33
Attitude Control	15
Margin(15%)	14.6
Residual(2%)	2.2
Total	113.8

DART (impact probe)

Element	Mass(kg)
Velocity Control	
- Orbit insertion	85
- Error correction	3
- Stationkeeping	5
- Station Change	7
Attitude Control	20
Margin(15%)	12
Residual(2%)	2.6
Total	134.6

3.3 Budgets for S/C Power

AIM (rendezvous probe)

Subsystems	Power (W)
Payloads	100
S/C bus	
- Propulsion	1000
- ADCS	20
- Comm	15
- C&DH	25
- Thermal	120
- Power	100
- Structure	-
Margin(15%)	207
Total	1587

DART (impact probe)

Subsystems	Power (W)
Payloads	29
S/C bus	
- Propulsion	8.8
- ADCS	26.4
- Comm	8.8
- C&DH	25
- Thermal	25
- Power	52.7
- Structure	-
Margin(15%)	26.3
Total	202

3.3 Budgets for S/C Weight

AIM (rendezvous probe)

Subsystems	Weight (kg)
Payloads	15
S/C bus	
- Propulsion	20
- ADCS	10
- Comm	30
- C&DH	14.5
- Thermal	30
- Power	20
- Structure	35

Subsystems	Weight (kg)
Sum of S/C bus	159.5
Margin (10%)	17.5
S/C Dry Weight	192
Propellant	113.8
Loaded Weight	305.8

3.3 Budgets for S/C Weight

DART (impact probe)

Subsystems	Weight (kg)
Payloads	5
S/C bus	
- Propulsion	10
- ADCS	10
- Comm	13
- C&DH	13
- Thermal	5
- Power	6.2
- Structure	15

Subsystems	Weight (kg)
Sum of S/C bus	72.2
Margin (30%)	23.2
S/C Dry Weight	100.4
Propellant	134.6
Loaded Weight	235

3.4 Preliminary S/S designs

① Propulsion S/S

AIM
(rendezvous
probe)

type	specifications
Kick Motor Thruster	STAR-48
Chemical Thruster	SNECMA PPS-1350 Hall-effect thruster; Xenon ion을 이용 The number of thrusters: 12
Electric Propulsion system	Solar Electric Propulsion (SEP); $I_{sp}=1000\sim7000s$

DART
(impact
probe)

type	specifications
Kick Motor Thruster	STAR-48
Chemical Thruster	Hydrazine(N_2H_4) Monopropellant; The number of thrusters: 12

3.4 Preliminary S/S designs

② ADCS

AIM(rendezvous probe)

Type	three-axis attitude control
Sensor	Sun Tracker (0.2~1kg, 0~0.2W)
number of sensors	2



DART(impact probe)

Type	three-axis attitude control
Sensor	Star Tracker (5~50kg, 2~20W)
number of sensors	2



3.4 Preliminary S/S designs

③ Comm

AIM(rendezvous probe)

Antenna	Fixed High-Gain Antenna
Frequency	X-band & S-band telemetry



DART(impact probe)

Antenna	Fixed High-Gain Antenna(직경 1m) 2 Low-Gain Antenna
Frequency	X-band telemetry
speed	1.4-3Mbps downlink, 2kbps uplink for HGA 50/125bps up/downlink for LGA



3.4 Preliminary S/S designs

④ C&DH

AIM(rendezvous probe)

Command unit	5.0kg, 5.4W on standby 14W on operating
Pulse Code Modulation Encoder	5.5kg, 5.5W

DART(impact probe)

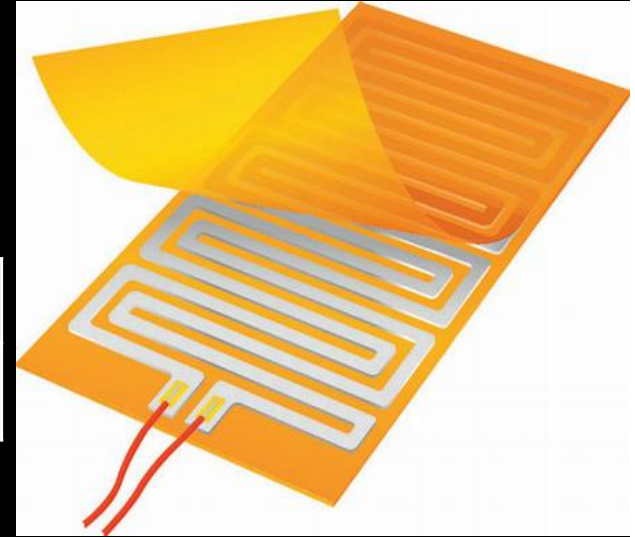
Command unit	5.0kg, 5.4W on standby 14W on operating
Pulse Code Modulation Encoder	5.5kg, 5.5W
Autonomous guidance algorithm based on APL experience in development of Standard Missile	

3.4 Preliminary S/S designs

⑤ Thermal S/S

AIM(rendezvous probe)

Active control	Heat Pipes (25kg, 100W)
Passive control	Coating



DART(impact probe)

Active control	Heater (0.3kg, 25W)
Passive control	Coating



3.4 Preliminary S/S designs

⑥ EPS

AIM(rendezvous probe)

Solar Array	Deployment after launch
Battery type	NiCd; 20~40W*hr/kg capacity

DART(impact probe)

Solar Array	2.7m ² ; Deployment after launch
Battery type	32V Lithium-ion; 300W*hr capacity for launch

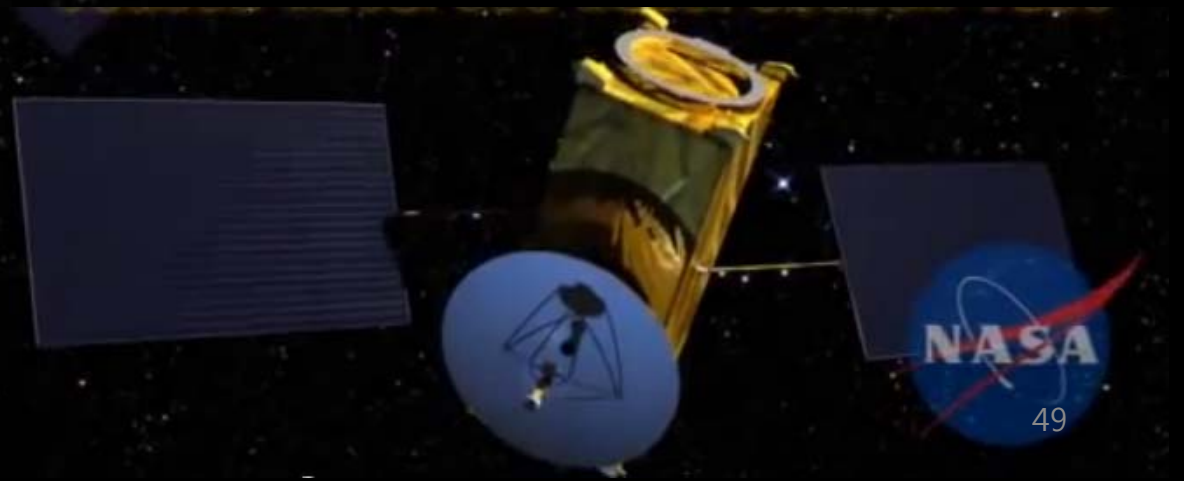
3.4 Preliminary S/S designs

⑦ Structure & Mechanisms

AIM(rendezvous probe)



DART(impact probe)



Thank you!

